Assignment 3 - Duke Lemur Center

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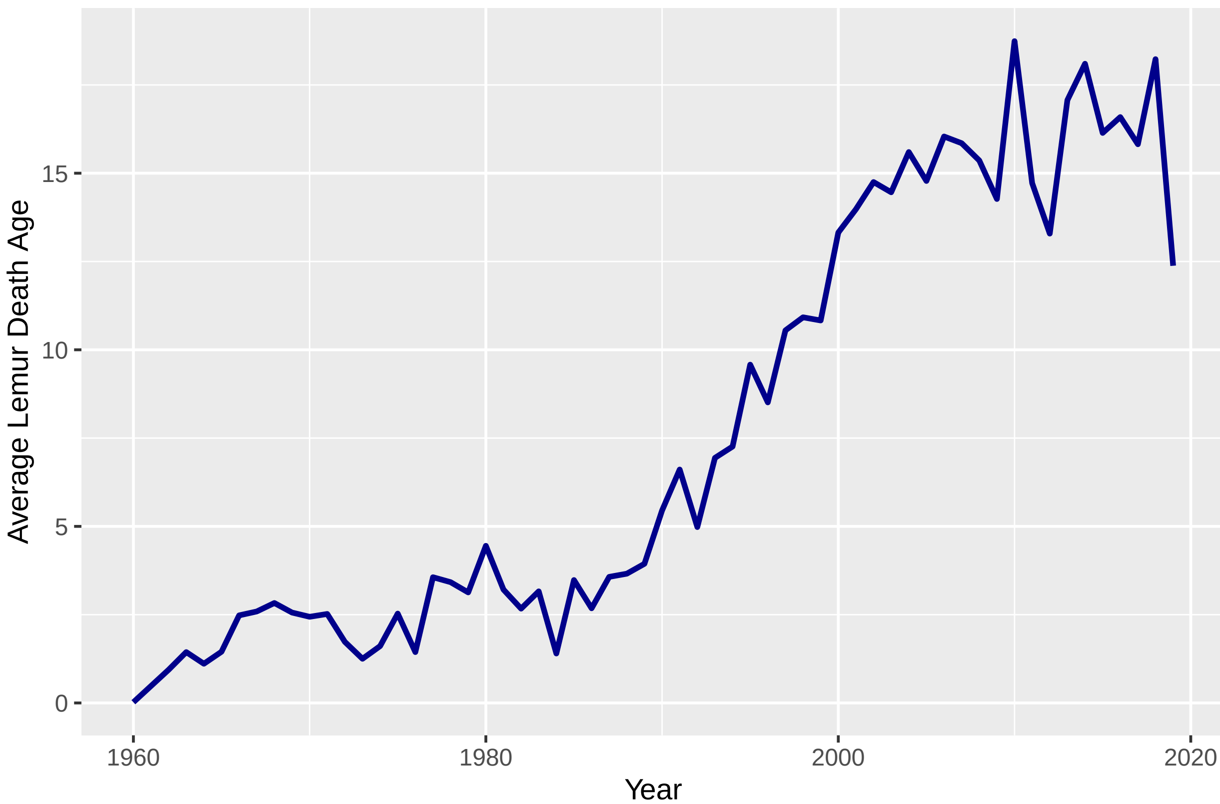

Figure 1. Duke Lemur Center residents' average death age from 1960 to 2019

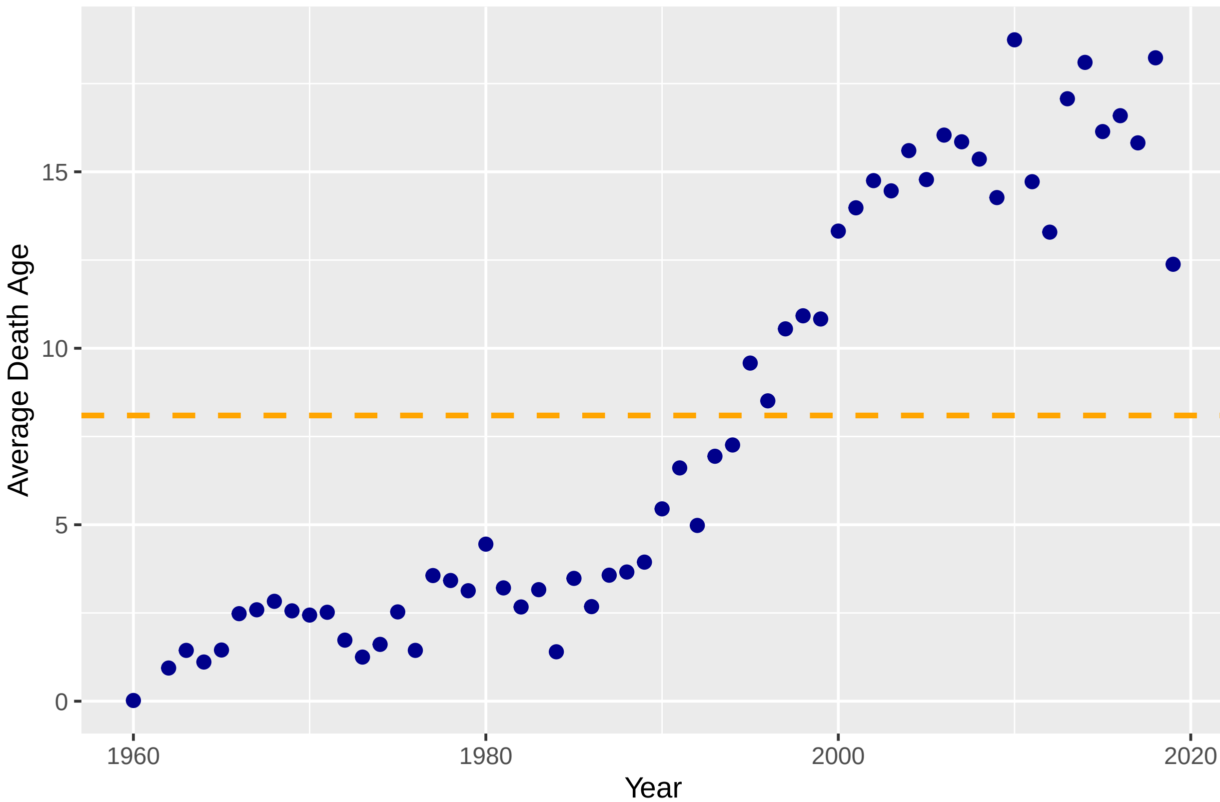

Figure 2. Duke Lemur Center residents' average death age from 1960 to 2019

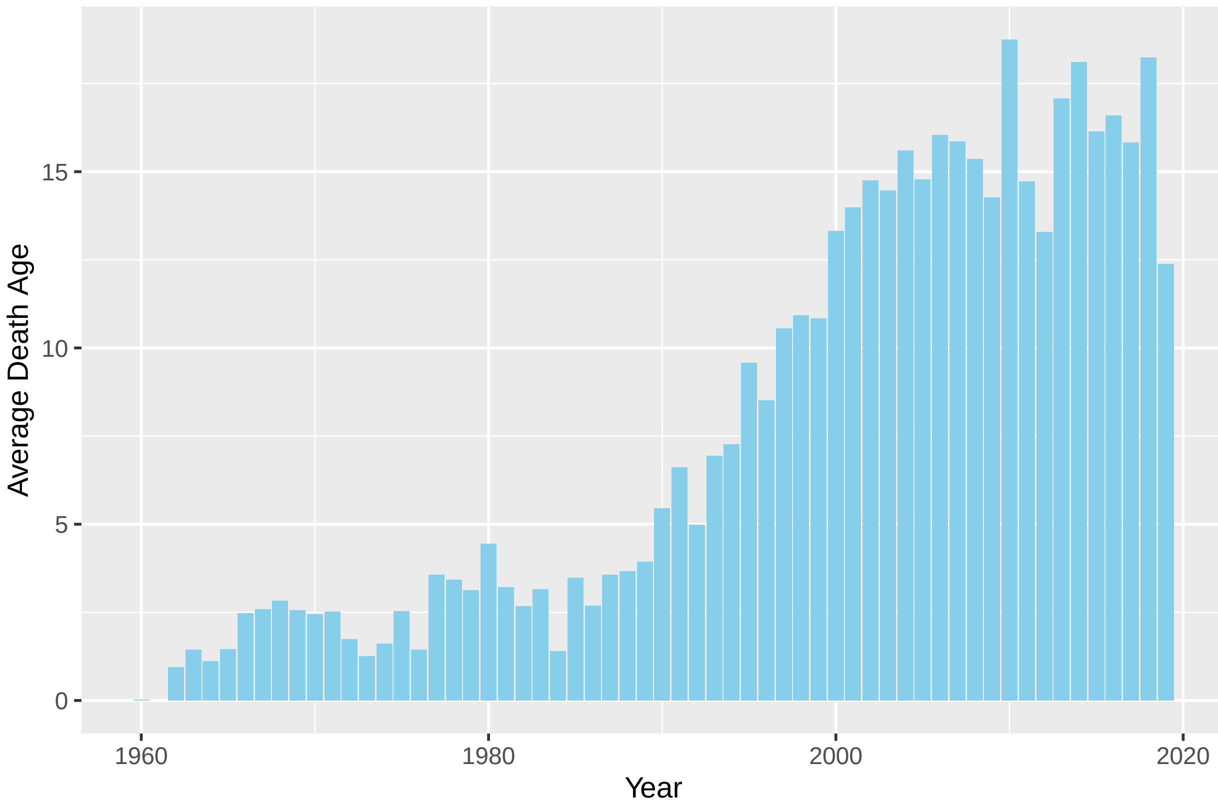

Figure 3. Duke Lemur Center residents' average death age from 1960 to 2019

```r
library(readxl)
DLCLH <- read_excel("~/Downloads/Duke Lemur Center Life History.xlsx")
library(ggplot2)
library(dplyr)
library(reshape2)
library(lubridate)
```

```r
unique(DLCLH$Taxon)

#AVG death age overall
mean(DLCLH$AgeAtDeath_y, na.rm = TRUE)

#this code is showing the mean death age for each taxon
avg_death <- DLCLH %>%
  group_by(Taxon) %>%
  summarize(Avg_AgeDeath=mean(AgeAtDeath_y, na.rm = TRUE))
```

```r
#this code will be to seperate out the death dates
death_date <- DLCLH %>%
  mutate(deathdate = as.POSIXct(DOD, format = "%Y-%m-%d")) %>%
  mutate(Year = year(deathdate)) %>%
  mutate(Month = month(deathdate)) %>%
  mutate(Day = day(deathdate))

unique(death_date$Year)

#This gives the min and max year dates recorded for deaths just to explore
mimaxyear <- death_date %>%
  summarize(min_year = min(Year, na.rm = TRUE),
            max_year = max(Year, na.rm = TRUE))
```

```r
#death average by year
death_avg <- death_date %>%
  group_by(Year) %>%
  summarize(Avg_AgeDeath=mean(AgeAtDeath_y, na.rm = TRUE),
            Max = max(AgeAtDeath_y, na.rm = TRUE))

#this will round Avg_AgeDeath so its easier to read
death_avg$Avg_AgeDeath <- round(death_avg$Avg_AgeDeath, digit = 2)


#this will be death average by taxon
#not sure how helpful this one is as it is so much information
tax_death <- death_date %>%
  group_by(Taxon, Year) %>%
  summarize(Avg_AgeDeath=mean(AgeAtDeath_y, na.rm = TRUE),
            Max = max(AgeAtDeath_y, na.rm = TRUE),
            Min = min(AgeAtDeath_y, na.rm = TRUE))
#so for now we can ignore
```

```r
#this is the death average by year and values are rounded
death_avg <- death_date %>%
  group_by(Year) %>%
  summarize(Avg_AgeDeath=mean(AgeAtDeath_y, na.rm = TRUE))
```

```r
death_avg$Avg_AgeDeath <- round(death_avg$Avg_AgeDeath, digit = 2)
#this is taking out the null values
death_avg <- death_avg %>% filter(!is.na(Year))
```

```r
#First plot is a line plot
ggplot(death_avg, aes(x= Year, y = Avg_AgeDeath)) +
  geom_line( stat = "identity", color = "darkblue", size = 1) +
  labs(x = "Year", y = "Average Lemur Death Age",
       caption = "Figure 1. Duke Lemur Center residents' average death age from 1960 to 2019")
```

```r
#Second plot
ggplot(death_avg, aes(x= Year, y = Avg_AgeDeath)) +
  geom_point(size = 2, color = "darkblue") +
  labs(x = "Year", y = "Average Death Age",
       caption = "Figure 2. Duke Lemur Center residents' average death age from 1960 to 2019") +
  geom_hline(yintercept = 8.095792, col = "orange", lty = 2, lwd = 1)
```

```r
#Third plot
ggplot(death_avg, aes(x= Year, y = Avg_AgeDeath)) +
  geom_bar( stat = "identity", fill = "skyblue") +
  labs(x = "Year", y = "Average Death Age",
       caption = "Figure 3. Duke Lemur Center residents' average death age from 1960 to 2019") +
  theme_gray()
```